Yongfeng Xu (Justin)

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EDUCATION

Nanyang Technological University (NTU), SG Aug 2025 - Present
- Electrical/Electronics & Computer Engineering, Expect MSc
- GPA: 4.33/5.0
- Data Mining, Machine Learning, Signal Analysis, Advanced Devices…

South China University of Technology (SCUT), CN Sep 2021 – Jun 2025
- Information Technology, BEng
- GPA: 3.44/4.0
- Signals & Systems, Python, C++, ML & DL Basics, Circuit Analysis (SLF, VLF, HF, RF)…

INTERNSHIP

Desay SV Dec 2024 – May 2025

Algorithm Engineer
- Documented technical designs and functional safety requirements in alignment with ISO 26262 standards, ensuring compliance throughout the software development lifecycle.
- Profiling C pipelines to reduce compute overhead, ensuring real-time execution within strict ECU memory and hardware constraints, cutting data curation time for model response from 1.45s to 1.3s(approx.) (~10%).
- Assisted in early production evaluation and offshore mass production transfer, ensured quality robustness.

China Telecom Jun 2024 - Aug/2024

Product Engineer
- Built and deployed a performance dashboard for multiple IoT devices, cutting the downtime by (~13%).
- Coordinated cross-functional reviews between hardware, software, operation teams, streamlined system upgrade priorities.
- Conduct technical and market research, obtain product feature trends and define workflows.

PROJECT & RESEARCH

HDB Resale Market Intelligence Mar 2026 – Aug 2026
- **https://github.com/Constructer2023/HDB_resale_market_analysis.git**
- Build an end-end market intelligence pipeline processing 100,000+ public HDB resale transactions (data.gov.sg / URA API), engineering features for floor level, proximity to MRTs, remaining lease…
- Develop a ML based model (include XGBoost, RandomForest) to predict HDB resale flat prices.
- Architect a relational database schema to store transaction data, optimize query speed for market reporting.
- Integrate geospatial libraries (GeoPandas) and make use of OneMap API (a popular government-based geo info API) to map transactional hotspots, feature enrichment and estate pricing.
- Design an interactive dashboard (Power BI) to track pricing trends across different towns, enable users to evaluate historical price per sqm and identify undervalued resale units.

US Equity Market Risk & Market Regime Analytics Jun 2026 - Present
- Develop a market regime detection model using Hidden Markov Models (HMM) and unsupervised clustering (K-Means / GMM) to classify US equity market states (low volatility bull, high volatility bear, crisis).
- Zoom in and engineer macro trends and market features to signal regime transitions and stress periods.
- Design risk interactive dashboard to display real-time factor risk exposure, correlation breakdowns.
- Make stress testing simulations using historical events (2008 GFC, 2020 Covid Shock, 2022 Inflation).

UWB Radar Car Scene Application Aug 2024 – Dec 2024
- Utilized Python (data cleaning libraries) for data preprocessing to remove noise and enhance data quality.
- Developed algorithms in Python and MATLAB, employing Scikit-learn for parameter optimization to detect living beings within vehicles.
- Enriched feature patterns and did data mining about the raw UWB radar signal.
- Embedded software programmed in C and C++ to ensure real-time algorithm performance on hardware.
- Conducted field tests with radar hardware on actual vehicles to validate algorithmic accuracy (include mimic multiple interference scenarios).

Multimodal Sensing Platform: Radar Gesture & Facial Recognition Feb 2024 – Jun 2024

- Processed FMCW radar raw IQ data using 2D-FFT and STFT to generate range-Doppler and micro-Doppler heatmaps for hand gesture classification.
- Engineered a lightweight neural network (2D CNN-LSTM) to classify temporal radar spectrogram sequences, achieving ~93% real-time classification accuracy on low-power embedded hardware.
- Fused multimodal sensor inputs (radar gesture triggering + camera-based face verification) to reduce camera active runtime and cut total device power consumption by 60%+.

Design of Intelligent Health Aid Agricultural Applet Sep 2023 - Nov2023

- Designed a user-friendly front-end interface using HTML, CSS, and JavaScript.
- Created a back-end database with MySQL for product data storage and retrieval.
- Integrated an AI module to enable health question-answering functionality. Linked the front and back ends through API connections to facilitate interactive, intelligent responses.

PATENT

Synchronous Demodulation Method and System for Passive RFID Tags Granted**,** Oct 2025

- China National Intellectual Property Administration
- No. ZL 2025 1 0060679.1

EXTRACURRICULAR ACTIVITIES

Google Data Analytics @ Coursera Dec 2024 – Feb 2025

- Data analysis with R programming

Campus UP @ NTU Aug 2025 - Present

- Organize the weekly event about the club.
- Connect to potential sponsors.

English Corner @ SCUT Mar 2023 – Jul 2024

- Organized regular English-speaking practice sessions
- Involved in planning and hosting large-scale events, and liaised with other associations, such as the International Students Union (ISU, 2025 class) to create a welcoming environment for language and cultural exchange.

Radio Association @ SCUT Oct 2021 – Jan 2023

- Organized the Wireless Cup competition and assisted with technical sharing sessions to advance radio technology.
- Collaborated to support educational and competitive initiatives, and encouraged participation in radio-related activities, advancing technical skills through competitions and workshops.

SanYuanLi Volunteer Association Committee Member @ Canton City Dec 2024 – Jun 2025

- Maintained and organized the daily operations and administrative stuff, led the workflows.
- Planning and scheduling personnel management for large-scale volunteer events.

HONOURS AND AWARDS

- National Undergraduate Training Program for Innovation and Entrepreneurship Finalization, CN Apr 2024
- Merit Award, Greater Bay Area Cup Financial Mathematical Modelling Competition Nov 2023
- Meritorious, The Mathematical Contest in Modelling, US May 2024
- First Prize, National Undergraduate Innovation and Entrepreneurship Competition Jul 2023
- Third Prize National Scholarship, CN Dec 2022

SKILLS AND INTERESTS

Skills: MySQL, Python, NumPy, Pandas, Matplotlib, Seaborn, sklearn, PyTorch, Transformer, Power BI, R, Matlab, Git.

Interests (skills learning & developing): Data Mining, Data Scraping, CV, NLP, LLMs, etc.